2.7 Brainteasers.

1. A flea is going between two points which are 100 inches apart by jumping (always in the same direction) either one inch or two inches at a time. How many different paths can the flea travel by?

2. I have a bag containing three pancakes: one golden on both sides, one burnt on both sides, and one golden on one side and burnt on the other. You shake the bag, draw a pancake at random, look at one side, and notice that it is golden. What is the probability that the other side is golden?

3. Alice and Bob are playing heads and tails, Alice tosses $n+1$ coins, Bob tosses n coins. The coins are fair. What is the probability that Alice will have strictly more heads than Bob?

4. Alice is in a restaurant trying to decide between three desserts. How can she choose one of three desserts with equal probability with the help of a fair coin? What if the coin is biased and the bias is unknown?

5. What is the expected number of times you must flip a fair coin until it lands on head? What if the coin is biased and lands on head with probability p?

6. What is the expected number of coin tosses of a fair coin in order to get two heads in a row? What if